

Abhishek Reddy Manam

Data Engineer · Azure Data Factory · Databricks · PySpark · Snowflake · dbt · Kafka · Airflow · AWS · GCP · Docker

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PROFESSIONAL SUMMARY

Data Engineer with 3+ years of hands-on experience designing, building, and maintaining production-grade data pipelines that turn raw, messy data into reliable, analytics-ready assets. Works across the full data lifecycle — ingestion, transformation at scale, orchestration, and delivery — with deep expertise in the Azure ecosystem: Azure Data Factory for dynamic metadata-driven pipeline design, Azure Databricks and PySpark for large-scale distributed transformation, Delta Lake and Medallion Architecture (Bronze, Silver, Gold) on Azure Data Lake Gen2, and Azure Synapse Analytics for serving business-ready data to Power BI. Beyond Azure, brings strong cross-platform experience — Snowflake and dbt for data warehousing and analytics engineering, Apache Kafka for real-time event streaming, Apache Airflow for orchestrating complex multi-step workflows, and Docker for consistent reproducible deployment. AWS and GCP round out cross-cloud requirements. Cares as much about data reliability as data engineering: automated quality validation, pipeline monitoring, data lineage documentation, and governance standards built into every system. M.S. Information Systems Technologies, Wilmington University, Newark, DE (GPA 3.6, expected May 2026).

TECHNICAL SKILLS

Azure Data Factory & Azure: Built parameterised metadata-driven ADF pipelines using Lookup, ForEach, and Copy Activities with JSON-driven configuration — making pipelines scalable across multiple data sources without code changes. 40+ linked services (SQL Server, Dataverse, REST APIs, SFTP, Azure Blob, ADLS Gen2, Snowflake). Self-Hosted Integration Runtime for on-premise connectivity. Event-driven and schedule-based triggers. Mapping Data Flows for complex transformation logic. Azure DevOps for CI/CD release pipeline deployment. Azure Synapse Analytics Serverless SQL Pool (OPENROWSET, External Tables, SQL Views with Managed Identity) connected to Power BI. Azure Key Vault for secrets management. Azure Monitor for pipeline health dashboards and alerting. Reduced source onboarding time from 3 days to under 2 hours through parameterised pipeline templates adopted as the team standard.

Azure Databricks & PySpark: Transformed data at scale using PySpark in Azure Databricks — implemented Medallion Architecture (Bronze, Silver, Gold) with Delta Lake on Azure Data Lake Gen2. Applied type casting, null handling, derived columns, conditional logic, and multi-source joins. Optimised Spark workloads through broadcast joins, predicate pushdown, Adaptive Query Execution, and partition tuning — reducing pipeline latency by 50%. Delta Lake for ACID-compliant storage: MERGE INTO for CDC and SCD patterns, Z-ORDER clustering, schema evolution, and time-travel for audit and recovery. Delta Live Tables for declarative pipeline development. Unity Catalog for centralised governance and lineage. Databricks Workflows for end-to-end orchestration with dependency management.

Snowflake & dbt: Led dbt development in Snowflake with staging, intermediate, and mart layer architecture — mandatory schema test coverage (not_null, unique, relationships, accepted_values), SCD Type 2 snapshots for 6 dimension tables, incremental materialisation reducing full-refresh runtime by 60%, and dbt documentation as the living data dictionary. 80+ production models owned end-to-end. CI/CD-gated deployment via GitHub Actions with dbt slim-CI (70% build time reduction). Established data quality standards that reduced production incidents by 40%. Snowflake warehouse right-sizing, clustering key design, query profile analysis, RBAC role hierarchy, and column-level dynamic data masking. Compute costs reduced approximately 30% through systematic METERING_HISTORY analysis and incremental conversion.

Apache Kafka & Real-Time Streaming: Built real-time streaming pipelines using Apache Kafka and PySpark Structured Streaming — processing 5M+ events per day with exactly-once semantics, watermarking for late-arriving events, windowed aggregations, and dead-letter queue handling for failed records. Kafka consumer group design, partition key strategy, and topic configuration for high-throughput low-latency event processing. Applied streaming output to downstream analytics consumption and

operational reporting. Azure Event Hubs (Kafka-compatible endpoint) for managed streaming on Azure. AWS Kinesis for streaming on AWS. Real-time pipeline health monitoring with structured alerting.

Apache Airflow & Orchestration: Owned Airflow DAG development for orchestrating complex multi-step workflows — 8+ production DAGs with dynamic task generation from configuration files (reducing DAG file count by 70%), TaskFlow API, SLA miss callbacks with PagerDuty and Slack alerting, retry-with-backoff for transient failures, backfill management, and idempotent task design. Cross-service DAGs spanning Databricks, ADF, Snowflake, S3, and REST API ingestion stages. Mean time to resolution reduced from 2 hours to 30 minutes through structured root-cause runbooks covering 15+ common failure patterns. AWS MWAA for managed Airflow on AWS.

Python & SQL: Python for ETL pipeline development — class-based OOP architecture with type hints, Pandas for tabular data manipulation, PySpark for distributed workloads, REST API clients with OAuth2 and pagination handling, SQLAlchemy and database drivers for multi-platform connectivity, and pytest for unit and integration test coverage on all pipeline components. SQL across Snowflake, Databricks SQL, Azure Synapse, PostgreSQL, SQL Server, BigQuery, and Amazon Redshift — CTEs, window functions, complex multi-table JOINS, stored procedures, execution plan analysis, index design, and query optimisation for analytical workloads. Shell scripting for pipeline automation and deployment tasks.

AWS, GCP & Docker: AWS for cross-cloud data engineering: S3 (partitioned data lake), Glue (managed ETL and data catalog), EMR (managed Spark clusters), Redshift (cloud warehouse), Lambda (serverless event-driven triggers), SQS (event queuing), CloudWatch (monitoring and alerting), IAM (least-privilege access). GCP: BigQuery (partitioned and clustered tables, cost optimization, INFORMATION_SCHEMA governance), Cloud Composer (managed Airflow), Cloud Storage, Dataflow (batch and stream processing). Docker for containerised pipeline deployment — multi-stage Dockerfiles, docker-compose for local development, ECR for container registry. Terraform for infrastructure as code across AWS and Azure. GitHub Actions CI/CD for automated testing and environment promotion.

Data Quality & Governance: Implemented data quality frameworks to catch anomalies before impacting downstream systems — Great Expectations validation suites (schema conformance, null rate thresholds, referential integrity, value range guards, distribution drift detection) as mandatory pipeline build gates, reducing quality incidents by 40%. dbt schema tests and source freshness checks as CI requirements. Cross-source reconciliation SQL for pipeline accuracy verification. Anomaly detection alerting for unexpected volume changes. MTTR reduced from 2 hours to 30 minutes through structured root-cause runbooks. Data lineage documentation through dbt ref() DAG and Confluence architecture diagrams. RBAC, column-level masking, encryption, and audit logging for governance and compliance. Power BI semantic models (DAX, DirectQuery, Row-Level Security) serving 200+ business users.

EXPERIENCE

Data Engineer • *Digiuniv Technologies Private Limited* • *Hyderabad, India (Hybrid)* Jun 2023 – Jul 2024

- Joined to design and scale the enterprise data platform from the ground up, connecting ERP, field operations, finance, and reporting systems into a trusted analytics-ready data layer. Architected the end-to-end Azure data platform — ADF metadata-driven pipelines with JSON-config parameterisation, Databricks PySpark Medallion Architecture (Bronze/Silver/Gold) on Delta Lake, and Synapse Serverless SQL views connected to Power BI — enabling the business to move from manual Excel reporting to automated dashboards. Reduced source onboarding time from days to hours by introducing parameterised ADF pipeline templates adopted as the team engineering standard for all new data source integrations.
- Designed and owned the dbt transformation layer in Snowflake — established staging/intermediate/mart architecture with mandatory schema test coverage and CI/CD-gated deployment via GitHub Actions, reducing production data quality incidents by 40%. Built the organisation's first real-time streaming pipeline using Apache Kafka and PySpark Structured Streaming, processing 5M+ events/day with exactly-once semantics for downstream analytics consumption. Optimised Spark workloads through broadcast joins, predicate pushdown, and partition tuning — reducing pipeline latency by 50%.
- Partnered with Finance, Operations, and Analytics teams to translate business reporting requirements into scalable data models, data dictionaries, and reusable Power BI semantic models serving 200+ business users. Implemented Airflow DAG orchestration with dynamic task generation and SLA monitoring. Applied data quality frameworks, lineage documentation, and governance standards across all pipeline assets. Mentored 2 junior engineers through weekly code reviews, architecture walkthroughs, and pull request feedback — contributing to team engineering standards documentation.

Associate Data Engineer • *Blockysite* • *Remote*

Mar 2022 – Jun 2023

- Stepped into a full-time role with ownership of designing, building, and maintaining data pipelines that powered analytics and business reporting. Built dynamic Azure Data Factory pipelines using Lookup, ForEach, and Copy Activities with JSON-driven configuration and parameterised datasets — making pipelines scalable across multiple data sources without code changes. Transformed data at scale using PySpark in Azure Databricks, implementing Medallion Architecture (Bronze, Silver, Gold) with Delta Lake on Azure Data Lake Gen2, applying type casting, null handling, derived columns, conditional logic, and multi-source joins.
- Served business-ready data through Azure Synapse Analytics Serverless SQL Pool using OPENROWSET, SQL Views, and External Tables with Managed Identity authentication, connected to Power BI for reporting. Built real-time streaming pipelines using Apache Kafka and PySpark, processing high-volume event data with low latency. Led dbt development in Snowflake with staging, intermediate, and mart layers including testing, incremental models, and documentation. Owned Airflow DAG development for orchestrating complex workflows with SLA monitoring and dynamic pipeline generation. Worked with AWS (S3, Glue, Redshift, EMR) for cross-cloud requirements. Containerised pipeline services using Docker with CI/CD integration. Implemented data quality frameworks and mentored junior engineers through pull request reviews.

Data Engineer Intern • Blockysite • Remote

Jun 2021 – Mar 2022

- Joined the data engineering team and got hands-on with the full data stack from raw ingestion through clean, query-ready tables. Contributed to Azure Data Factory pipeline development, PySpark transformations on Databricks, SQL query development, data quality checks, and documentation. Built the engineering foundations — clean code, testing discipline, and collaborative review habits — that carried into every subsequent role.

S E L E C T E D P R O J E C T S

Enterprise Azure Data Platform — End-to-End Pipeline Architecture | *Azure Data Factory • Azure Databricks • PySpark • Delta Lake • Azure Synapse Analytics • Snowflake • dbt • Power BI • Apache Kafka • Apache Airflow • Docker • GitHub Actions*

- Architected and delivered an enterprise data platform connecting 40+ source systems to a centralised analytics layer — ADF metadata-driven ingestion framework (JSON-config parameterisation, 40+ linked services — 85% deployment effort reduction, onboarding 3 days to under 2 hours), Azure Databricks PySpark Medallion Architecture on Delta Lake (50% latency reduction through Spark optimisation), dbt Snowflake transformation layer (80+ models, schema tests, SCD Type 2 for 6 dimensions, incremental materialisation — 60% runtime reduction, 40% quality incident reduction), and Apache Kafka + PySpark Structured Streaming (5M+ events/day, exactly-once semantics).
- Airflow orchestration (8+ production DAGs, dynamic generation — 70% DAG file reduction, SLA monitoring — MTTR 2 hours to 30 minutes), Azure Synapse Serverless SQL Pool connected to Power BI (200+ business users, DAX semantic models, Row-Level Security), Snowflake RBAC and cost governance (~30% compute reduction), Docker containerisation, GitHub Actions CI/CD (dbt slim-CI — 70% build time reduction), Great Expectations data quality gates (40% incident reduction), and comprehensive lineage documentation. Mentored 2 junior engineers through code reviews and architecture guidance.

E D U C A T I O N

M.S. Information Systems Technologies • Wilmington University, Newark, DE • GPA 3.6 / 4.0 *Aug 2024 – May 2026 (Expected)*

B.Tech. Computer Science & Engineering • Lovely Professional University, Punjab, India